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also consumes some power. The backlight type LCD display and the MCU also consume some amount of power.

Considering the battery being charged to be lead-acid type (battery type not mentioned anywhere in the article), trickle-charging is achieved only with no-load condition of the battery. As per the circuit on page 104, the battery is always loaded!

For other types of batteries like NiCd, trickle charging is achieved by reducing the charging current. According to the circuit, the relay just connects the battery to the source when the solar panel produces sufficient voltage and it cannot control the charging current needed for trickle charging.

2. As per the circuit, there is no provision for measurement of charging current. For measurement of power/energy, both voltage and current have to be measured. Also, the article does not describe as to how the energy and power are measured.

If no current measurement is done, a solar panel of the same voltage rating but with different current capacity will display the same result, which is not possible.

Vivek Ghosh
West Bengal

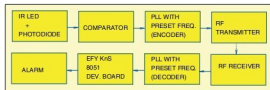
The author Santhosh Jayarajan replies:

1. Trickle charge is not achieved by controlling the current but by switching the load 'on' and 'off' very much like PWM. The circuit does consume power from the battery but this loss of battery power is compensated by the solar panel.

Readers should go through the program to see that in trickle mode, there are sequential 'on' and 'off' cycles. The timer interrupt takes care of all the timing for 'on'/'off' control. It also provides a 1-second interrupt to do the energy integration. It should be remembered that in this case, the energy will be a summation of the power every second.

I have never said that the current would change during trickle charge. The circuit is meant for a lead-acid battery.

2. The low-value resistor in the



Block diagram for anti-petrol theft alarm

circuit will drop a voltage depending on the current flow. This voltage is measured and scaled for current. A PIC16F877 can measure only voltage, not current, at its input analogue pins. PIC16F877 datasheet should be referred for more information.

The program should be gone through in detail to understand how it works. For better understanding, one should have basic knowledge about timer interrupts, PWM and calculation of energy by integration of power over a time period.

Anti-Petrol Theft Alarm

I am a third-year electronics engineering student. I want to undertake 'Anti-petrol Theft Alarm' published in EFY Sept. 2012 as my minor project. Please explain this system with a block diagram.

Ashish Nanaware
Through e-mail

EFY: Figure above shows the block diagram of anti-petrol theft alarm. The transmitter has to be mounted in the vehicle near the petrol tap. When someone comes near the petrol tap, the IR sensor (passive sensor) senses his presence and a voltage difference appears at IC LM741 (used as comparator). The output of the comparator goes high and transistor BC547 conducts, enabling IC PLL567 (used as encoder) to transmit a prefix frequency via RF transmitter.

At the receiving end, the transmitted frequency is decoded by IC PLL567 if the correct frequency is received. The output of the phase-locked loop (PLL) is fed to the input port of EFY-KnS 8051 development board (you can use 89V51RPL directly). On receiving the correct frequency, the output of the PLL goes low, which is detected at the MCU port. Now you can trigger any device that you want to by suitably modifying the code. In this project we have triggered a piezobuzzer.